

Total No. of Questions : 10]

SEAT No. :

P2415

[4758]-583

[Total No. of Pages : 3

T.E. (Computer Engineering)

EMBEDDED OPERATING SYSTEMS

(2012 Course) (Semester - II) (310250) (End-Sem.)

Time : 2.½ Hours]

[Max. Marks :70

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*
- 4) Assume suitable data, if necessary.*

Q1) a) With the help of a diagram explain the classification of real-time scheduling methods. **[6]**

b) Explain role of barrel shifter in the ARM. **[4]**

OR

Q2) a) Explain how does user space applications communicates with the hardware? **[4]**

b) Write an ARM assembly program to find value of expression $3X + 6Y + 9Z$, where $X = 4$, $Y = 5$, $Z = 3$. **[6]**

Q3) a) Write short notes on (any two): **[6]**

i) LSB

ii) OSDL

iii) Kernel command line processing.

b) Why Embedded Linux is popular as OS for embedded system development? **[4]**

OR

P.T.O.

- Q4)** a) What are the steps involved in ‘subsystem initialization’. [4]
b) Explain the steps involved in initialization flow of control on embedded Linux. [6]

- Q5)** a) Explain the role of bootloader in embedded systems. [4]
b) Give the general steps involved in PCI discovery process and probe function. [7]
c) Enlist device driver module utilities and explain the use of same. [6]

OR

- Q6)** a) What are the challenges for bootloader in embedded system? [7]
b) Explain the JFFS2 file system. [5]
c) Explain about U-boot configurable commands. [5]
- Q7)** a) Explain the tracing and profiling tools used in embedded application development. [6]
b) Write short note on (any two): [6]
i) GStreamer Media framework.
ii) OpenGL ES.
iii) ssh
c) How to debug the kernel using ‘printk’? [5]

OR

- Q8)** a) Draw & explain KGDB setup for kernel debugging. [5]
b) Write short note on (any two): [6]
i) DDD
ii) EGL
iii) Open GL
c) Explain the working of stepper motor? State any two applications of stepper motor in embedded systems. [6]

- Q9)** a) Give the latency components in Linux with neat diagram. [8]
- b) Explain in detail, development process of Android applications. [8]

OR

- Q10)** a) What policies are used by Linux to schedule a real time task? [8]
- b) Write short notes on (any two): [8]
- i) Dalvik VM
 - ii) Zygote
 - iii) Activity Manager

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